AI Pilot

RESEARCH DOCUMENTATION

LEARNING, MEMORY & STRUCTURAL ADAPTATION

Adaptive Llama Research Results

Changing-world learning, held-out forgetting, compute-budgeted comparisons and an inspectable C++ demonstrator.

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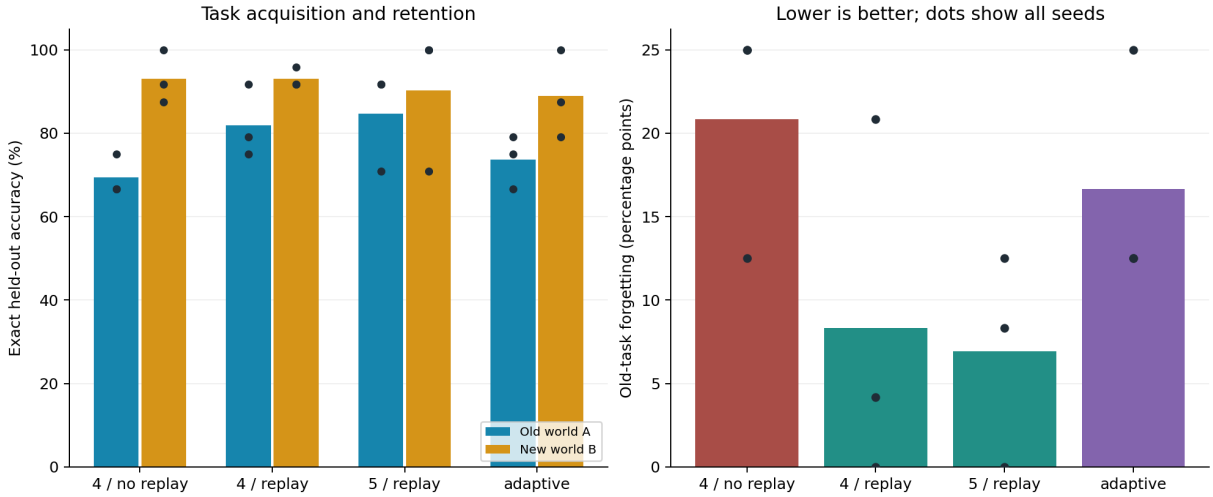
Adaptive Llama: changing rules, memory and structural growth

Measured 11 October 2026. Native C++ / LibTorch CPU experiment, with independent llama.cpp F16 deployment checks. This report completes the requested two-world experiment, forgetting measurement, fixed/adaptive comparison and live viewer. The model predicts coordinates; it is not a conversational assistant.

Development and verification of this bounded demonstrator are complete and the experiment is closed. Native resume/export and browser checks passed. The documented model errors and lack of adaptive advantage are measured outcomes; they were not removed by post-hoc retuning. Alternative-architecture development is a separate next task.

Result

Under this declared toy-task protocol, adding 25% old-world replay to the fixed four-layer learner reduced mean forgetting by **12.50 percentage points**. The adaptive method proposed growth in 3/3 seeds and accepted it in 0/3. Its mean new-world test accuracy differed from the fixed four-layer replay control by **-4.17 percentage points**. These results do not establish a useful accuracy advantage from structural growth.



Mean held-out accuracy and forgetting; dots show the three individual seeds.

Mean results across the three seeds

Condition	Old A accuracy	New B accuracy	Changed B only	Forgetting
Fixed 4, no replay	69.44%	93.06%	94.44%	20.83 pp
Fixed 4, replay	81.94%	93.06%	83.33%	8.33 pp
Fixed 5, replay	84.72%	90.28%	94.44%	6.94 pp
Adaptive, replay	73.61%	88.89%	77.78%	16.67 pp

Each task has 24 held-out test combinations. The changed-answer subset contains 6 combinations per world. Old accuracy is after adaptation; forgetting is old accuracy before minus after the switch. Negative forgetting means improvement. The dots and per-seed table expose variation; three seeds are not enough for a general superiority claim.

All measured runs

Seed / condition	A before	A after	B before	B after	Forgetting
42 / Fixed 4, no replay	91.7%	66.7%	66.7%	91.7%	25.0 pp
42 / Fixed 4, replay	91.7%	91.7%	66.7%	91.7%	0.0 pp
42 / Fixed 5, replay	91.7%	91.7%	66.7%	100.0%	0.0 pp
42 / Adaptive, replay	91.7%	79.2%	66.7%	79.2%	12.5 pp
43 / Fixed 4, no replay	100.0%	75.0%	75.0%	100.0%	25.0 pp
43 / Fixed 4, replay	100.0%	79.2%	75.0%	95.8%	20.8 pp
43 / Fixed 5, replay	100.0%	91.7%	79.2%	100.0%	8.3 pp
43 / Adaptive, replay	100.0%	75.0%	75.0%	100.0%	25.0 pp
44 / Fixed 4, no replay	79.2%	66.7%	70.8%	87.5%	12.5 pp
44 / Fixed 4, replay	79.2%	75.0%	70.8%	91.7%	4.2 pp
44 / Fixed 5, replay	83.3%	70.8%	79.2%	70.8%	12.5 pp
44 / Adaptive, replay	79.2%	66.7%	70.8%	87.5%	12.5 pp

What the model was asked to learn

In world A, R/L/U/D move right/left/up/down on a 4x4 grid. World B reverses those four directions. Wait, obstacles and boundary clamping remain unchanged. For example, `w=0,x=1,y=2,a=R,b=0>` maps to `2,2;`, while the same input under `w=1` maps to `0,2;`. The world ID is supplied explicitly. This tests retaining and selecting contextual rules; it does not test discovering an unannounced change.

The original 160 combinations keep their fixed 112/24/24 train/validation/test partition in both worlds. No held-out combination is trained under the other world ID. Blocked and wait cases often have unchanged answers, so changed-answer accuracy is reported separately. All output generation is unconstrained; wrong or malformed coordinates count as errors.

The model uses 128 hidden dimensions, four attention heads, a fixed byte vocabulary, RMSNorm, RoPE and SwiGLU. Four layers contain 787,584 parameters; five contain 951,680. Training is in C++ using the existing LibTorch installation. The replay buffer contains only old training examples, never old test examples.

Comparison and compute budget

Four conditions use seeds 42, 43 and 44. Four-layer variants share exactly the same pretrained checkpoint per seed. Fixed five layers are trained from the start under the same matrix-work budget. Replay batches contain exactly 12 new and four old examples; the no-replay control uses 16 new examples.

Pretraining has a 600-four-layer-update budget; adaptation has 400. The declared compute proxy is three times the forward dense-matrix multiply-add count:

```
batch * sequence * (width*vocabulary + layers*(4*width^2 + 3*width*FF + 2*sequence*width)).
```

The factor of three approximates forward plus backward computation. It excludes nonlinearities, normalization, Adam, validation, checkpointing and export. Thus these are approximately matched matrix-work budgets, not exact FLOPs or equal wall-clock budgets. Integer updates can leave a small unused remainder.

Condition	Pretrain updates	Adaptation updates, all branches	Proxy used	Training CPU time
Fixed 4, no replay	600	400	100.00%	19.97 s
Fixed 4, replay	600	400	100.00%	22.85 s
Fixed 5, replay	488	325	99.77%	24.98 s
Adaptive, replay	600	372	99.95%	23.76 s

CPU times above include all adaptation training branches, including rejected candidates; they exclude validation/export and pretraining. Measurements ran serially with two CPU threads on a shared Intel i5-1335U laptop. Other workloads remained active. Wall time and CPU time differ; no isolated speed advantage is claimed. Full per-run timing and matrix-work values are in the JSON evidence.

The final campaign invocation took 290.17 s wall time and 577.95 s process CPU time, with 193.48 MiB peak RSS. It resumed after a deliberate pause at initial update 100; these invocation totals exclude that earlier process. The segment counters include its training. Shared pretraining is performed once for the three four-layer conditions, so summing attributed per-condition pretraining times overcounts actual campaign work.

Structural decision and its cost

The adaptive learner first receives 100 new-world updates with replay. If new validation accuracy is below 98%, loss above 0.02, or old validation accuracy has dropped more than two points, it proposes a fifth identity-initialized block. Otherwise it spends the remaining budget continuing four layers.

When proposed, the remaining matrix-work budget is split between the continued four-layer control and the five-layer candidate. Both start from the same warm checkpoint and sampling state. The larger candidate gets fewer updates; both branches are charged to the adaptive method. As a result, the selected branch receives fewer updates than a fixed model that spends its entire budget on one trajectory. This is an explicit search cost, not hidden extra compute.

Acceptance requires at least 1% lower new-world validation loss, non-regressing new accuracy, and old validation accuracy no worse than the control and within two percentage points of the pre-switch reference. Final-test metrics never enter this decision. The rule is written by us, not learned by the model.

Seed	Proposed / accepted	Control B loss	Candidate B loss	Control / candidate A accuracy
42	yes / no	0.27067	0.10128	58.3% / 87.5%

Seed	Proposed / accepted	Control B loss	Candidate B loss	Control / candidate A accuracy
43	yes / no	0.07015	0.10731	66.7% / 70.8%
44	yes / no	0.25393	0.26804	75.0% / 75.0%

What can be concluded

Replay and structural growth are separate mechanisms. In this setup, retaining old examples has a measurable benefit, but replay does not guarantee zero forgetting in every seed. The fixed-five-layer control helps separate extra capacity from an adaptive search procedure. Extra capacity and a successful layer insertion do not by themselves establish a useful adaptation policy.

A concrete limitation appeared in seed 42. The five-layer candidate improved new-world validation loss from 0.27067 to 0.10128 and old-world validation accuracy from 58.33% to 87.50% relative to its continued control. However, the pre-switch old-world validation reference was 91.67%; a 4.17-point loss exceeds the predeclared two-point guard. The rule therefore rejected this candidate and chose a control with worse retention. **The fallback is not a retention guarantee.** With only 24 validation examples, even one error costs 4.17 points, so the two-point threshold effectively allows no lost correct example. This shows a limitation of the selection rule, not proof that every growth candidate is worse. We did not relax the rule after seeing the results or evaluate rejected candidates on the test set.

This experiment is small, uses a supplied task ID, and has only 6 changed-answer test combinations and three random seeds. Results are sensitive to initialization, training budget, replay ratio and the decision rule. The study was not tuned against the final test outcomes. It does not establish a new learning paradigm, open-ended self-improvement, natural-language ability or frontier intelligence.

Live and offline inspection

The live viewer is at http://127.0.0.1:8090/ while its Node server runs. It reads atomic status snapshots from the C++ trainer, checks once per second and stops polling when the campaign is complete. The trainer measures after each 50-update chunk. A short highlight means a new measurement arrived; it is not an animation of individual neuron activation. The history slider can inspect a real candidate with five layers and compare its old/new validation curves.

The versioned offline report is `experiments/adaptive-llama/examples/continual.html`. Open it directly in a browser; no login, model or server is required. Final test comparisons are separated from the validation timeline. The machine-readable snapshot is `continual.json` alongside it.

From the repository root, start a new run manually:

```
./experiments/adaptive-llama/tools/run.sh campaign \
  artifacts/adaptive-llama/my-campaign
```

In a second terminal, serve its live measurements:

```
node experiments/adaptive-llama/tools/live.cjs \
  artifacts/adaptive-llama/my-campaign
```

Resume an interrupted campaign with `run.sh campaign-resume DIRECTORY`. Keep all run files together. The native learner, Adam, random state and segment timing are checkpointed atomically. Completed conditions and baseline checkpoints are reused. Stop the viewer separately with Ctrl+C. No boot/login service is created.

Try the model in the official llama.cpp chat

Start the existing llama.cpp built-in web UI with the trained snapshot:

```
./experiments/adaptive-llama/tools/start-chat.sh
```

Open http://127.0.0.1:8088/ and start a new conversation. The model name is `Adaptive-Llama-Toy`; the default snapshot is seed 42, adaptive with replay. This is the same llama.cpp interface used for Qwen, now serving the tiny model. The separate measurement viewer stays at port 8090.

Send the following strings exactly, without Markdown backticks:

Message	Expected	Observed model answer
`w=0,x=1,y=2,a=R,b=0>`	`2,2;`	`0,2;` (wrong)
`w=1,x=1,y=2,a=R,b=0>`	`0,2;`	`0,2;` (correct)
`w=1,x=1,y=2,a=R,b=1>`	`1,2;`	`1,2;` (correct)

The first error illustrates old-world forgetting. The expected answers come from the world rules; the chat displays only the model's generated answer, without a simulator replacing it. Checks use greedy generation and four tokens.

The custom chat template sends only the latest user message, stripped of outer whitespace. System messages and previous turns are absent from the predictor's input. This provides no conversational memory, chat learning or natural-language competence. Use a text-only coordinate prompt. The visible chat history is a UI record. Do not attach files.

Defaults are temperature 0, top-k 1, repeat penalty 1 and max tokens 4. Previous browser preferences may override UI defaults: reset those fields if necessary. The model is held in RAM until Ctrl+C stops the server. This manual server holds the adaptive experiment lock; stop it before training. Qwen uses the same port: stop the existing server before switching. An optional first launcher argument selects another generated GGUF snapshot. No boot/login service is installed.

Configuration authority: the experiment README records the exact selected GGUF, its SHA-256, model alias, local addresses, runtime bounds and observed responses. The versioned launcher is `experiments/adaptive-llama/tools/start-chat.sh`; the chat template and UI defaults are `ui/last-user.jinja` and `ui/llama-settings.json` in the same experiment directory. Changes to these settings must update that record. The source files live in Git; generated models remain in ignored artifacts.

Deployment and verification

All 12 selected models were exported in F32 and F16 GGUF. Native/llama.cpp logits passed the existing 0.0003 F32 and 0.02 F16 maximum-error thresholds. F16 llama.cpp exact accuracy matched native accuracy on both test worlds and the changed-answer subset. A four-layer F16 file is 1,592,576 bytes; five layers are 1,921,824 bytes.

The GGUF is a frozen predictor. The adaptation rule, replay and optimizer are in the C++ learner/native checkpoints, not executable logic inside GGUF. The complete campaign retains approximately 408 MiB of local checkpoints and exports, despite the small size of each inference model. These are generated artifacts, not extra model downloads, and are excluded from Git.

The separate verification manifest records source/artifact hashes and native, resume, UI and document checks. A live test observed an actual 150-to-200 update transition, and the browser supports recorded four/five-layer history inspection.

Efficiency review: task examples remain in memory; metrics are measured only at bounded checkpoints; status JSON is written once per observation. The browser uses one read-only status request per second and stops after completion. The viewer serves only its page and status, on loopback. Export checks recompute short prefixes for correctness; they are not an optimized serving benchmark.

Next research boundary

The separate alternative-architecture project has not been implemented by this campaign. The user supplied TypeSafe AI's Jev/System One Models as a reference for that next discussion. Its architecture, access and claims need separate assessment; Jev is not the JEPA research line. No Jev API or model was used here.